**1) a)**

$$\begin{pmatrix}1 & 0 & -4\\ 0 & 2 & -5\\ 0 & 0 & -3\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\lambda\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}1 & 0 & -4\\ 0 & 2 & -5\\ 0 & 0 & -3\end{pmatrix}-\lambda\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}1-\lambda & 0 & -4\\ 0 & 2-\lambda & -5\\ 0 & 0 & -3-\lambda\end{pmatrix}$$

$\det(\mathbf{A}-\lambda\mathbf{I})=(1-\lambda)(2-\lambda)(-3-\lambda)-0+0$

So $(1-\lambda)(2-\lambda)(-3-\lambda)=0$

$\Rightarrow \lambda_1=-3,\ \lambda_2=1$ or $\lambda_3=2$

$\therefore \lambda_3=2$

**1) b)**

When $\lambda_3=2$ then $\mathbf{A}-2\mathbf{I}=\begin{pmatrix}-1 & 0 & -4\\ 0 & 0 & -5\\ 0 & 0 & -5\end{pmatrix}$

$$\begin{pmatrix}-1 & 0 & -4\\ 0 & 0 & -5\\ 0 & 0 & -5\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=0$$

$\Rightarrow -x-4z=0 \quad \therefore x=-4z$

$-5z=0 \quad \therefore z=0$ so that $x=z=0$

$\therefore \mathbf{v}_3=\begin{pmatrix}0\\ 1\\ 0\end{pmatrix}$

$$\mathbf{P}=\begin{pmatrix}1 & 1 & 0\\ 1 & 0 & 1\\ 1 & 0 & 0\end{pmatrix}$$

$$\Rightarrow \det\mathbf{P}=\begin{vmatrix}1 & 1 & 0\\ 1 & 0 & 1\\ 1 & 0 & 0\end{vmatrix}=-1(-1)+0-0=1$$

Cofactor matrix of $\mathbf{P}$: $\begin{pmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & -1 & -1\end{pmatrix}$

$$\therefore \mathbf{P}^{-1}=\begin{pmatrix}0 & 0 & 1\\ 1 & 0 & -1\\ 0 & 1 & -1\end{pmatrix}$$

**1) c)**

$$\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$$

$$= \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 & -4 \\ 0 & 2 & -5 \\ 0 & 0 & -3 \end{pmatrix}\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}\begin{pmatrix} -3 & 1 & 0 \\ -3 & 0 & 2 \\ -3 & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} -3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

**2) a)**

$$\begin{pmatrix}5 & 1 & -5\\ 2 & 3 & -4\\ 2 & 1 & -2\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\lambda\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}5 & 1 & -5\\ 2 & 3 & -4\\ 2 & 1 & -2\end{pmatrix}-\lambda\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}5-\lambda & 1 & -5\\ 2 & 3-\lambda & -4\\ 2 & 1 & -2-\lambda\end{pmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I})=\begin{vmatrix}5-\lambda & 1 & -5\\ 2 & 3-\lambda & -4\\ 2 & 1 & -2-\lambda\end{vmatrix}$$

$$=\begin{vmatrix}5-\lambda & 1 & -5\\ 0 & 2-\lambda & -2+\lambda\\ 2 & 1 & -2-\lambda\end{vmatrix}$$

$$=\begin{vmatrix}5-\lambda & 1 & -4\\ 0 & 2-\lambda & 0\\ 2 & 1 & -1-\lambda\end{vmatrix}$$

$$=\begin{vmatrix}3-\lambda & 0 & -3+\lambda\\ 0 & 2-\lambda & 0\\ 2 & 1 & -1-\lambda\end{vmatrix}$$

$$=(3-\lambda)\big((2-\lambda)(-1-\lambda)\big)-2(-3+\lambda)(2-\lambda)$$

$$=(3-\lambda)(2-\lambda)(-1-\lambda+2)$$

$$=(3-\lambda)(2-\lambda)(1-\lambda)$$

So $(3-\lambda)(2-\lambda)(1-\lambda)=0$

$\therefore \lambda_1=1$, $\lambda_2=2$ and $\lambda_3=3$

When $\lambda_1=1$ then $\mathbf{A}-1\mathbf{I}=\begin{pmatrix}4 & 1 & -5\\ 2 & 2 & -4\\ 2 & 1 & -3\end{pmatrix}$

$$\begin{pmatrix}4 & 1 & -5\\ 2 & 2 & -4\\ 2 & 1 & -3\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=0$$

$\Rightarrow 4x+y-5z=0$ (1)

$2x+2y-4z=0$ (2)

$2x+y-3z=0$ (3)

(1) − (2): $2x-2z=0$ $\therefore x=z$

Sub into (3): $y-x=0$ $\therefore y=x$ $\quad\therefore \mathbf{v}_1=\begin{pmatrix}1\\ 1\\ 1\end{pmatrix}$

When $\lambda_2=2$ then $\mathbf{A}-2\mathbf{I}=\begin{pmatrix}3 & 1 & -5\\ 2 & 1 & -4\\ 2 & 1 & -4\end{pmatrix}$

$$\begin{pmatrix}3 & 1 & -5\\ 2 & 1 & -4\\ 2 & 1 & -4\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=0$$

$\Rightarrow 3x+y-5z=0$ (1)

$2x+y-4z=0$ (2)

$2x+y-4z=0$ (3)

(1) − (2): $x-z=0 \therefore x=z$

Sub into (2): $y-2z=0 \ \therefore y=2z \ \ \therefore \mathbf{v}_2=\begin{pmatrix}1\\ 2\\ 1\end{pmatrix}$

When $\lambda_3=3$ then $\mathbf{A}-3\mathbf{I}=\begin{pmatrix}2 & 1 & -5\\ 2 & 0 & -4\\ 2 & 1 & -5\end{pmatrix}$

$$\begin{pmatrix}2 & 1 & -5\\ 2 & 0 & -4\\ 2 & 1 & -5\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=0$$

$\Rightarrow 2x+y-5z=0$ (1)

$2x-4z=0$ (2)

$2x+y-5z=0$ (3)

From (2): $x-2z=0 \ \therefore x=2z$

Sub into (1): $y-z=0 \ \therefore y=z \qquad \therefore \mathbf{v}_3=\begin{pmatrix}2\\ 1\\ 1\end{pmatrix}$

$$\therefore \mathbf{P}=\begin{pmatrix}1 & 1 & 2\\ 1 & 2 & 1\\ 1 & 1 & 1\end{pmatrix}$$

$$\mathbf{P}^{-1}=\begin{pmatrix}-1 & -1 & 3\\ 0 & 1 & -1\\ 1 & 0 & -1\end{pmatrix}$$

$\mathbf{D}=\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$

$$=\begin{pmatrix}-1 & -1 & 3\\ 0 & 1 & -1\\ 1 & 0 & -1\end{pmatrix}\begin{pmatrix}5 & 1 & -5\\ 2 & 3 & -4\\ 2 & 1 & -2\end{pmatrix}\begin{pmatrix}1 & 1 & 2\\ 1 & 2 & 1\\ 1 & 1 & 1\end{pmatrix}=\begin{pmatrix}-1 & -1 & 3\\ 0 & 1 & -1\\ 1 & 0 & -1\end{pmatrix}\begin{pmatrix}1 & 2 & 6\\ 1 & 4 & 3\\ 1 & 2 & 3\end{pmatrix}$$

$$=\begin{pmatrix}1 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 3\end{pmatrix}$$

$$\therefore \mathbf{P}=\begin{pmatrix}1 & 1 & 2\\ 1 & 2 & 1\\ 1 & 1 & 1\end{pmatrix} \text{ and } \mathbf{D}=\begin{pmatrix}1 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 3\end{pmatrix}$$

**2) b)**

$$\begin{pmatrix}1 & 2 & 1\\ 2 & 3 & 1\\ 1 & 4 & 3\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\lambda\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}1 & 2 & 1\\ 2 & 3 & 1\\ 1 & 4 & 3\end{pmatrix}-\lambda\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}1-\lambda & 2 & 1\\ 2 & 3-\lambda & 1\\ 1 & 4 & 3-\lambda\end{pmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I})=\begin{vmatrix}1-\lambda & 2 & 1\\ 2 & 3-\lambda & 1\\ 1 & 4 & 3-\lambda\end{vmatrix}$$

$$=\begin{vmatrix}1-\lambda & 2 & 1\\ 2 & 3-\lambda & 1\\ -1+2\lambda & 0 & 1-\lambda\end{vmatrix}$$

$$=\begin{vmatrix}1-\lambda & 2 & 1\\ 1+\lambda & 1-\lambda & 0\\ -1+2\lambda & 0 & 1-\lambda\end{vmatrix}$$

$$=1\big(0-(1-\lambda)(-1+2\lambda)\big)-0+(1-\lambda)\big((1-\lambda)(1-\lambda)-2(1+\lambda)\big)$$

$$=-(1-\lambda)(-1+2\lambda)+(1-\lambda)(\lambda^2-4\lambda-1)$$

$$=(1-\lambda)(1-2\lambda+\lambda^2-4\lambda-1)$$

$$=(1-\lambda)(\lambda^2-6\lambda)$$

$$=\lambda(1-\lambda)(\lambda-6)$$

So $\lambda(1-\lambda)(\lambda-6)=0$

$\therefore \lambda_1=0$, $\lambda_2=1$ and $\lambda_3=6$

When $\lambda_1=0$ then $\mathbf{A}-0\mathbf{I}=\begin{pmatrix}1 & 2 & 1\\ 2 & 3 & 1\\ 1 & 4 & 3\end{pmatrix}$

$$\begin{pmatrix}1 & 2 & 1\\ 2 & 3 & 1\\ 1 & 4 & 3\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=0$$

$\Rightarrow x+2y+z=0$ (1)

$2x+3y+z=0$ (2)

$x+4y+3z=0$ (3)

(3) − (1): $2y+2z=0$ $\therefore z=-y$

Sub into (1): $x+y=0$ $\therefore -y=x$ $\therefore \mathbf{v}_1=\begin{pmatrix}1\\ -1\\ 1\end{pmatrix}$

When $\lambda_2=1$ then $\mathbf{A}-1\mathbf{I}=\begin{pmatrix}0 & 2 & 1\\ 2 & 2 & 1\\ 1 & 4 & 2\end{pmatrix}$

$$\begin{pmatrix}0 & 2 & 1\\ 2 & 2 & 1\\ 1 & 4 & 2\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=0$$

$\Rightarrow 2y+z=0$ (1)

$2x+2y+z=0$ (2)

$x+4y+2z=0$ (3)

From (1) $2y+z=0 \therefore -z=2y$

Sub into (2): $2x=0 \ \therefore x=0 \ \therefore \mathbf{v}_2=\begin{pmatrix}0\\ 1\\ -2\end{pmatrix}$

When $\lambda_3=6$ then $\mathbf{A}-6\mathbf{I}=\begin{pmatrix}1 & 2 & 1\\ 2 & 3 & 1\\ 1 & 4 & 3\end{pmatrix}$

$$\begin{pmatrix}-5 & 2 & 1\\ 2 & -3 & 1\\ 1 & 4 & -3\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=0$$

$\Rightarrow -5x+2y+z=0$ (1)

$2x-3y+z=0$ (2)

$x+4y-3z=0$ (3)

(1) − (2): $-7x+5y+z=0 \ \therefore 7x=5y$

$\Rightarrow \frac{7}{5}x=y$

Sub into (3): $\frac{33}{5}x-3z=0 \ \therefore \frac{11}{5}x=z \qquad \therefore \mathbf{v}_3=\begin{pmatrix}5\\ 7\\ 11\end{pmatrix}$

$$\therefore \mathbf{P}=\begin{pmatrix}1 & 0 & 5\\ -1 & 1 & 7\\ 1 & -2 & 11\end{pmatrix}$$

$$\mathbf{P}^{-1}=\frac{1}{30}\begin{pmatrix}25 & -10 & -5\\ 18 & 6 & -12\\ 1 & 2 & 1\end{pmatrix}$$

$$\mathbf{D}=\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$$

$$=\frac{1}{30}\begin{pmatrix}25 & -10 & -5\\ 18 & 6 & -12\\ 1 & 2 & 1\end{pmatrix}\begin{pmatrix}1 & 2 & 1\\ 2 & 3 & 1\\ 1 & 4 & 3\end{pmatrix}\begin{pmatrix}1 & 0 & 5\\ -1 & 1 & 7\\ 1 & -2 & 11\end{pmatrix}$$

$$=\frac{1}{30}\begin{pmatrix}25 & -10 & -5\\ 18 & 6 & -12\\ 1 & 2 & 1\end{pmatrix}\begin{pmatrix}0 & 0 & 30\\ 0 & 1 & 42\\ 0 & -2 & 66\end{pmatrix}$$

$$=\frac{1}{30}\begin{pmatrix}0 & 0 & 0\\ 0 & 30 & 0\\ 0 & 0 & 180\end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 6 \end{pmatrix}$$

$$\therefore \mathbf{P} = \begin{pmatrix} 1 & 0 & 5 \\ -1 & 1 & 7 \\ 1 & -2 & 11 \end{pmatrix} \text{ and } \mathbf{D} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 6 \end{pmatrix}$$

**3) a)**

$$\begin{pmatrix}2 & 1 & 0\\ 0 & 2 & 0\\ 0 & 0 & 3\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\lambda\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}2 & 1 & 0\\ 0 & 2 & 0\\ 0 & 0 & 3\end{pmatrix}-\lambda\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}2-\lambda & 1 & 0\\ 0 & 2-\lambda & 0\\ 0 & 0 & 3-\lambda\end{pmatrix}$$

$\det(\mathbf{A}-\lambda\mathbf{I})=(2-\lambda)(2-\lambda)(3-\lambda)$

So $(2-\lambda)(2-\lambda)(3-\lambda)=0$

$\therefore \lambda_1=2$, $\lambda_2=3$ and $\lambda_3=3$

When $\lambda_1=2$ then $\mathbf{A}-2\mathbf{I}=\begin{pmatrix}0 & 1 & 0\\ 0 & 0 & 0\\ 0 & 0 & 1\end{pmatrix}$

$$\begin{pmatrix}0 & 1 & 0\\ 0 & 0 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=0$$

$\Rightarrow y=0$ (1)

$z=0$ (3)

$$\therefore \mathbf{v}_1=\begin{pmatrix}1\\ 0\\ 0\end{pmatrix}$$

When $\lambda_2=\lambda_3=3$ then $\mathbf{A}-3\mathbf{I}=\begin{pmatrix}-1 & 1 & 0\\ 0 & -1 & 0\\ 0 & 0 & 0\end{pmatrix}$

$$\begin{pmatrix}-1 & 1 & 0\\ 0 & -1 & 0\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=0$$

$\Rightarrow -x+y=0$ (1)

$-y=0$ (2)

$\therefore x=y=0$

$$\therefore \mathbf{v}_2=\mathbf{v}_3=\begin{pmatrix}0\\ 0\\ 1\end{pmatrix}$$

$$\therefore \lambda_1=2,\ \mathbf{v}_1=\begin{pmatrix}1\\ 0\\ 0\end{pmatrix} \text{ and } \lambda_2=\lambda_3=3,\ \mathbf{v}_2=\mathbf{v}_3=\begin{pmatrix}0\\ 0\\ 1\end{pmatrix}$$

**3) b)**

We need 3 independent eigenvectors for a $3\times 3$ matrix to be diagonalisable but we have only 2

**4) a)**

$$\begin{pmatrix}2 & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & 1\end{pmatrix}^6 = \begin{pmatrix}2^6 & 0 & 0\\ 0 & (-1)^6 & 0\\ 0 & 0 & 1^4\end{pmatrix} = \begin{pmatrix}64 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}$$

**4) b)**

$$\begin{pmatrix}\frac{1}{2} & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & -2\end{pmatrix}^9 = \begin{pmatrix}\left(\frac{1}{2}\right)^9 & 0 & 0\\ 0 & (-1)^9 & 0\\ 0 & 0 & (-2)^9\end{pmatrix} = \begin{pmatrix}\frac{1}{512} & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & -512\end{pmatrix}$$

**4) c)**

$$\begin{pmatrix}a & 0 & 0\\ 0 & b & 0\\ 0 & 0 & c\end{pmatrix}^k = \begin{pmatrix}a^k & 0 & 0\\ 0 & b^k & 0\\ 0 & 0 & c^k\end{pmatrix}$$

**5)**

$\mathbf{P} = \begin{pmatrix}1 & 2 & 0\\ 0 & 1 & 1\\ 1 & -1 & 3\end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix}-1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 3\end{pmatrix}$

$\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} \Rightarrow \mathbf{P}\mathbf{D} = \mathbf{A}\mathbf{P} \Rightarrow \mathbf{P}\mathbf{D}\mathbf{P}^{-1} = \mathbf{A} \therefore \mathbf{A}^4 = \mathbf{P}\mathbf{D}^4\mathbf{P}^{-1}$

$$\mathbf{P}^{-1} = \frac{1}{6}\begin{pmatrix}4 & -6 & 2\\ 1 & 3 & -1\\ -1 & 3 & 1\end{pmatrix}$$

$$\mathbf{A}^4 = \mathbf{P}\mathbf{D}^4\mathbf{P}^{-1} = \frac{1}{6}\begin{pmatrix}1 & 2 & 0\\ 0 & 1 & 1\\ 1 & -1 & 3\end{pmatrix}\begin{pmatrix}-1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 3\end{pmatrix}^4\begin{pmatrix}4 & -6 & 2\\ 1 & 3 & -1\\ -1 & 3 & 1\end{pmatrix}$$

$$= \frac{1}{6}\begin{pmatrix}1 & 2 & 0\\ 0 & 1 & 1\\ 1 & -1 & 3\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 81\end{pmatrix}\begin{pmatrix}4 & -6 & 2\\ 1 & 3 & -1\\ -1 & 3 & 1\end{pmatrix}$$

$$= \frac{1}{6}\begin{pmatrix}1 & 2 & 0\\ 0 & 1 & 1\\ 1 & -1 & 3\end{pmatrix}\begin{pmatrix}4 & -6 & 2\\ 1 & 3 & -1\\ -81 & 243 & 81\end{pmatrix}$$

$$= \frac{1}{6}\begin{pmatrix}6 & 0 & 0\\ -80 & 246 & 80\\ -240 & 720 & 246\end{pmatrix}$$

$$= \frac{1}{3}\begin{pmatrix}3 & 0 & 0\\ -40 & 123 & 40\\ -120 & 360 & 123\end{pmatrix}$$

**6)**

$\mathbf{P} = \begin{pmatrix} 0 & 1 & 1 \\ 2 & 1 & 0 \\ 1 & 1 & 3 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} -5 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$\mathbf{D} = \mathbf{P}^{-1}\mathbf{MP} \Rightarrow \mathbf{PD} = \mathbf{MP} \Rightarrow \mathbf{PDP}^{-1} = \mathbf{M} \therefore \mathbf{M}^{10} = \mathbf{PD}^{10}\mathbf{P}^{-1}$

$\mathbf{P}^{-1} = \frac{1}{5}\begin{pmatrix} -3 & 2 & 1 \\ 6 & 1 & -2 \\ -1 & -1 & 2 \end{pmatrix}$

$\mathbf{M}^{10} = \mathbf{PD}^{10}\mathbf{P}^{-1} = \frac{1}{5}\begin{pmatrix} 0 & 1 & 1 \\ 2 & 1 & 0 \\ 1 & 1 & 3 \end{pmatrix}\begin{pmatrix} -5 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{10}\begin{pmatrix} -3 & 2 & 1 \\ 6 & 1 & -2 \\ -1 & -1 & 2 \end{pmatrix}$

$= \mathbf{\frac{1}{5}}\begin{pmatrix} 0 & 1 & 1 \\ 2 & 1 & 0 \\ 1 & 1 & 3 \end{pmatrix}\begin{pmatrix} 5^{10} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} -3 & 2 & 1 \\ 6 & 1 & -2 \\ -1 & -1 & 2 \end{pmatrix}$

$= \mathbf{\frac{1}{5}}\begin{pmatrix} 0 & 1 & 1 \\ 2 & 1 & 0 \\ 1 & 1 & 3 \end{pmatrix}\begin{pmatrix} -3 \times 5^{10} & 2 \times 5^{10} & 5^{10} \\ 6 & 1 & -2 \\ -1 & -1 & 2 \end{pmatrix}$

$= \mathbf{\frac{1}{5}}\begin{pmatrix} 5 & 0 & 0 \\ -6 \times 5^{10} + 6 & 4 \times 5^{10} + 1 & 2 \times 5^{10} - 2 \\ -3 \times 5^{10} + 3 & 2 \times 5^{10} - 2 & 5^{10} + 4 \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 & 0 \\ -6 \times 5^{9} + \frac{6}{5} & 4 \times 5^{9} + \frac{1}{5} & 2 \times 5^{9} - \frac{2}{5} \\ -3 \times 5^{9} + \frac{3}{5} & 2 \times 5^{9} - \frac{2}{5} & 5^{9} + \frac{4}{5} \end{pmatrix}$

**7) a)**

$$\mathbf{P}=\begin{pmatrix}1&0&1\\0&1&1\\2&0&1\end{pmatrix} \text{ and } \mathbf{D}=\begin{pmatrix}\frac{1}{2}&0&0\\0&\frac{1}{3}&0\\0&0&1\end{pmatrix}$$

$$\mathbf{D}=\mathbf{P}^{-1}\mathbf{M}\mathbf{P}\Rightarrow\mathbf{P}\mathbf{D}=\mathbf{M}\mathbf{P}\Rightarrow\mathbf{P}\mathbf{D}\mathbf{P}^{-1}=\mathbf{M}\therefore\mathbf{M}^n=\mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$$

$$\mathbf{P}^{-1}=\begin{pmatrix}-1&0&1\\-2&1&1\\2&0&-1\end{pmatrix}$$

$$\mathbf{M}^n=\mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}=\begin{pmatrix}1&0&1\\0&1&1\\2&0&1\end{pmatrix}\begin{pmatrix}\frac{1}{2}&0&0\\0&\frac{1}{3}&0\\0&0&1\end{pmatrix}^n\begin{pmatrix}-1&0&1\\-2&1&1\\2&0&-1\end{pmatrix}$$

$$=\begin{pmatrix}1&0&1\\0&1&1\\2&0&1\end{pmatrix}\begin{pmatrix}\left(\frac{1}{2}\right)^n&0&0\\0&\left(\frac{1}{3}\right)^n&0\\0&0&1\end{pmatrix}\begin{pmatrix}-1&0&1\\-2&1&1\\2&0&-1\end{pmatrix}$$

$$=\begin{pmatrix}1&0&1\\0&1&1\\2&0&1\end{pmatrix}\begin{pmatrix}-\left(\frac{1}{2}\right)^n&0&\left(\frac{1}{2}\right)^n\\-2\left(\frac{1}{3}\right)^n&\left(\frac{1}{3}\right)^n&\left(\frac{1}{3}\right)^n\\2&0&-1\end{pmatrix}$$

$$=\begin{pmatrix}2-\left(\frac{1}{2}\right)^n&0&\left(\frac{1}{2}\right)^n-1\\2-2\left(\frac{1}{3}\right)^n&\left(\frac{1}{3}\right)^n&\left(\frac{1}{3}\right)^n-1\\2-2\left(\frac{1}{2}\right)^n&0&2\left(\frac{1}{2}\right)^n-1\end{pmatrix}$$

$$=\begin{pmatrix}2-\frac{1}{2^n}&0&\frac{1}{2^n}-1\\2-\frac{2}{3^n}&\frac{1}{3^n}&\frac{1}{3^n}-1\\2-\frac{2}{2^n}&0&\frac{2}{2^n}-1\end{pmatrix}$$

**7) b)**

$$\mathbf{L}=\lim_{n\to\infty}\mathbf{M}^n=\lim_{n\to\infty}\begin{pmatrix}2-\frac{1}{2^n}&0&\frac{1}{2^n}-1\\2-\frac{2}{3^n}&\frac{1}{3^n}&\frac{1}{3^n}-1\\2-\frac{2}{2^n}&0&\frac{2}{2^n}-1\end{pmatrix}=\begin{pmatrix}2&0&-1\\2&0&-1\\2&0&-1\end{pmatrix}$$

**8) a)**

$$\mathbf{P}=\begin{pmatrix}1 & 2 & 1\\ 2 & -1 & 1\\ 1 & 1 & 2\end{pmatrix} \text{ and } \mathbf{D}=\begin{pmatrix}1 & 0 & 0\\ 0 & \frac{3}{4} & 0\\ 0 & 0 & \frac{1}{5}\end{pmatrix}$$

$$\mathbf{D}=\mathbf{P}^{-1}\mathbf{M}\mathbf{P}\Rightarrow\mathbf{P}\mathbf{D}=\mathbf{M}\mathbf{P}\Rightarrow\mathbf{P}\mathbf{D}\mathbf{P}^{-1}=\mathbf{M}\therefore\mathbf{M}^{n}=\mathbf{P}\mathbf{D}^{n}\mathbf{P}^{-1}$$

$$\mathbf{P}^{-1}=\frac{1}{6}\begin{pmatrix}3 & 3 & -3\\ 3 & -1 & -1\\ -3 & -1 & 5\end{pmatrix}$$

$$\mathbf{M}^{n}=\mathbf{P}\mathbf{D}^{n}\mathbf{P}^{-1}=\frac{1}{6}\begin{pmatrix}1 & 2 & 1\\ 2 & -1 & 1\\ 1 & 1 & 2\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 0 & \frac{3}{4} & 0\\ 0 & 0 & \frac{1}{5}\end{pmatrix}^{n}\begin{pmatrix}3 & 3 & -3\\ 3 & -1 & -1\\ -3 & -1 & 5\end{pmatrix}$$

$$=\frac{1}{6}\begin{pmatrix}1 & 2 & 1\\ 2 & -1 & 1\\ 1 & 1 & 2\end{pmatrix}\begin{pmatrix}1^{n} & 0 & 0\\ 0 & \left(\frac{3}{4}\right)^{n} & 0\\ 0 & 0 & \frac{1}{5^{n}}\end{pmatrix}\begin{pmatrix}3 & 3 & -3\\ 3 & -1 & -1\\ -3 & -1 & 5\end{pmatrix}$$

$$=\frac{1}{6}\begin{pmatrix}1 & 2 & 1\\ 2 & -1 & 1\\ 1 & 1 & 2\end{pmatrix}\begin{pmatrix}3 & 3 & -3\\ 3\left(\frac{3}{4}\right)^{n} & -\left(\frac{3}{4}\right)^{n} & -\left(\frac{3}{4}\right)^{n}\\ -\frac{3}{5^{n}} & -\frac{1}{5^{n}} & \frac{5}{5^{n}}\end{pmatrix}$$

$$=\frac{1}{6}\begin{pmatrix}3+6\left(\frac{3}{4}\right)^{n}-\frac{3}{5^{n}} & 3-2\left(\frac{3}{4}\right)^{n}-\frac{1}{5^{n}} & -3-2\left(\frac{3}{4}\right)^{n}-\frac{3}{5^{n}}\\ 6-3\left(\frac{3}{4}\right)^{n}-\frac{3}{5^{n}} & 6+\left(\frac{3}{4}\right)^{n}-\frac{1}{5^{n}} & -6+\left(\frac{3}{4}\right)^{n}-\frac{3}{5^{n}}\\ 3+3\left(\frac{3}{4}\right)^{n}-\frac{6}{5^{n}} & 3-\left(\frac{3}{4}\right)^{n}-\frac{2}{5^{n}} & -3-\left(\frac{3}{4}\right)^{n}-\frac{6}{5^{n}}\end{pmatrix}$$

**8) b)**

$$\mathbf{L}=\lim_{n\to\infty}\mathbf{M}^{n}=\frac{1}{6}\begin{pmatrix}3 & 3 & -3\\ 6 & 6 & -6\\ 3 & 3 & -3\end{pmatrix}=\frac{1}{2}\begin{pmatrix}1 & 1 & -1\\ 2 & 2 & -2\\ 1 & 1 & -1\end{pmatrix}$$

**9)**

$$\begin{pmatrix} k & 0 & k \\ 0 & k & 0 \\ k & 0 & k \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} k & 0 & k \\ 0 & k & 0 \\ k & 0 & k \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} k-\lambda & 0 & k \\ 0 & k-\lambda & 0 \\ k & 0 & k-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (k-\lambda)(k-\lambda)(k-\lambda) - 0 + k\big(-k(k-\lambda)\big)$$

$$= (k-\lambda)\big((k-\lambda)(k-\lambda) - k^2\big)$$

$$= (k-\lambda)(k^2 - 2k\lambda + \lambda^2 - k^2)$$

$$= \lambda(k-\lambda)(-2k+\lambda)$$

So $\lambda(k-\lambda)(\lambda-2k) = 0$

$\therefore \lambda_1 = 0$, $\lambda_2 = k$ and $\lambda_3 = 2k$

When $\lambda_1 = 0$ then $\mathbf{A} - 0\mathbf{I} = \begin{pmatrix} k & 0 & k \\ 0 & k & 0 \\ k & 0 & k \end{pmatrix}$

$$\begin{pmatrix} k & 0 & k \\ 0 & k & 0 \\ k & 0 & k \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow kx + kz = 0$ (1)

$ky = 0$ (2)

So $z = -x$, $y = 0$ $\therefore \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$

When $\lambda_2 = k$ then $\mathbf{A} - k\mathbf{I} = \begin{pmatrix} 0 & 0 & k \\ 0 & 0 & 0 \\ k & 0 & 0 \end{pmatrix}$

$$\begin{pmatrix} 0 & 0 & k \\ 0 & 0 & 0 \\ k & 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow kz = 0$ (1)

$kx = 0$ (2)

So $x = 0$ and $z = 0$ $\therefore \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

When $\lambda_3 = 2k$ then $\mathbf{A} - k\mathbf{I} = \begin{pmatrix} -k & 0 & k \\ 0 & -k & 0 \\ k & 0 & -k \end{pmatrix}$

$$\begin{pmatrix} -k & 0 & k \\ 0 & -k & 0 \\ k & 0 & -k \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow -kx + kz = 0 \qquad (1)$

$-ky = 0 \qquad (2)$

$\Rightarrow x = z,\ y = 0 \therefore \mathbf{v}_3 = \begin{pmatrix}1\\0\\1\end{pmatrix}$

$\mathbf{P} = \begin{pmatrix}1 & 0 & 1\\0 & 1 & 0\\-1 & 0 & 1\end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix}0 & 0 & 0\\0 & k & 0\\0 & 0 & 2k\end{pmatrix}$

$\mathbf{D} = \mathbf{P}^{-1}\mathbf{MP} \Rightarrow \mathbf{PD} = \mathbf{MP} \Rightarrow \mathbf{PDP}^{-1} = \mathbf{M} \therefore \mathbf{M}^n = \mathbf{PD}^n\mathbf{P}^{-1}$

$\mathbf{P}^{-1} = \frac{1}{2}\begin{pmatrix}1 & 0 & -1\\0 & 2 & 0\\1 & 0 & 1\end{pmatrix}$

$\mathbf{M}^n = \mathbf{PD}^n\mathbf{P}^{-1} = \frac{1}{2}\begin{pmatrix}1 & 0 & 1\\0 & 1 & 0\\-1 & 0 & 1\end{pmatrix}\begin{pmatrix}0 & 0 & 0\\0 & k & 0\\0 & 0 & 2k\end{pmatrix}^n\begin{pmatrix}1 & 0 & -1\\0 & 2 & 0\\1 & 0 & 1\end{pmatrix}$

$= \frac{1}{2}\begin{pmatrix}1 & 0 & 1\\0 & 1 & 0\\-1 & 0 & 1\end{pmatrix}\begin{pmatrix}0 & 0 & 0\\0 & k^n & 0\\0 & 0 & 2^n k^n\end{pmatrix}\begin{pmatrix}1 & 0 & -1\\0 & 2 & 0\\1 & 0 & 1\end{pmatrix}$

$= \frac{1}{2}\begin{pmatrix}1 & 0 & 1\\0 & 1 & 0\\-1 & 0 & 1\end{pmatrix}\begin{pmatrix}0 & 0 & 0\\0 & 2k^n & 0\\2^n k^n & 0 & 2^n k^n\end{pmatrix}$

$= \frac{1}{2}\begin{pmatrix}2^n k^n & 0 & 2^n k^n\\0 & 2k^n & 0\\2^n k^n & 0 & 2^n k^n\end{pmatrix}$

$= \frac{1}{2}k^n\begin{pmatrix}2^n & 0 & 2^n\\0 & 2 & 0\\2^n & 0 & 2^n\end{pmatrix}$

$= k^n\begin{pmatrix}2^{n-1} & 0 & 2^{n-1}\\0 & 1 & 0\\2^{n-1} & 0 & 2^{n-1}\end{pmatrix}$ as req.

**10) a)**

$$\begin{pmatrix} a-\frac{b}{2} & \frac{b}{2} & -a+\frac{b}{2} \\ a-b & b & -a+b \\ -\frac{b}{2} & \frac{b}{2} & \frac{b}{2} \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I} = \begin{pmatrix} a-\frac{b}{2} & \frac{b}{2} & -a+\frac{b}{2} \\ a-b & b & -a+b \\ -\frac{b}{2} & \frac{b}{2} & \frac{b}{2} \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} a-\frac{b}{2}-\lambda & \frac{b}{2} & -a+\frac{b}{2} \\ a-b & b-\lambda & -a+b \\ -\frac{b}{2} & \frac{b}{2} & \frac{b}{2}-\lambda \end{pmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I}) = \begin{vmatrix} a-\frac{b}{2}-\lambda & \frac{b}{2} & -a+\frac{b}{2} \\ a-b & b-\lambda & -a+b \\ -\frac{b}{2} & \frac{b}{2} & \frac{b}{2}-\lambda \end{vmatrix}$$

$$= \begin{vmatrix} a-\lambda & \frac{b}{2} & -a+\frac{b}{2} \\ a-\lambda & b-\lambda & -a+b \\ 0 & \frac{b}{2} & \frac{b}{2}-\lambda \end{vmatrix}$$

$$= \begin{vmatrix} a-\lambda & \frac{b}{2} & -a+\frac{b}{2} \\ 0 & \frac{b}{2}-\lambda & \frac{b}{2} \\ 0 & \frac{b}{2} & \frac{b}{2}-\lambda \end{vmatrix}$$

$$= \begin{vmatrix} a-\lambda & \frac{b}{2} & -a+\frac{b}{2} \\ 0 & -\lambda & \lambda \\ 0 & \frac{b}{2} & \frac{b}{2}-\lambda \end{vmatrix}$$

$$= \begin{vmatrix} a-\lambda & \frac{b}{2} & -a \\ 0 & -\lambda & 2\lambda \\ 0 & \frac{b}{2} & -\lambda \end{vmatrix}$$

$$= (a-\lambda)(\lambda^2-b\lambda)$$

$$= \lambda(a-\lambda)(\lambda-b)$$

$\therefore \lambda_1 = 0$, $\lambda_2 = a$ and $\lambda_3 = b$

**10) b)**

When $\lambda_1 = 0$ then $\mathbf{A} - 0\mathbf{I} = \begin{pmatrix} a - \frac{b}{2} & \frac{b}{2} & -a + \frac{b}{2} \\ a - b & b & -a + b \\ -\frac{b}{2} & \frac{b}{2} & \frac{b}{2} \end{pmatrix}$

$$\begin{pmatrix} a - \frac{b}{2} & \frac{b}{2} & -a + \frac{b}{2} \\ a - b & b & -a + b \\ -\frac{b}{2} & \frac{b}{2} & \frac{b}{2} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow \left(a - \frac{b}{2}\right)x + \left(\frac{b}{2}\right)y + \left(-a + \frac{b}{2}\right)z = 0$ (1)

$(a - b)x + by + (-a + b)z = 0$ (2)

$-\frac{b}{2}x + \frac{b}{2}y + \frac{b}{2}z = 0$ (3)

(1) − (3): $ax - az = 0 \therefore x = z$

Sub into (2); $by = 0 \therefore y = 0$

So $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

When $\lambda_2 = a$ then $\mathbf{A} - a\mathbf{I} = \begin{pmatrix} -\frac{b}{2} & \frac{b}{2} & -a + \frac{b}{2} \\ a - b & -a + b & -a + b \\ -\frac{b}{2} & \frac{b}{2} & -a + \frac{b}{2} \end{pmatrix}$

$$\begin{pmatrix} -\frac{b}{2} & \frac{b}{2} & -a + \frac{b}{2} \\ a - b & -a + b & -a + b \\ -\frac{b}{2} & \frac{b}{2} & -a + \frac{b}{2} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow -\frac{b}{2}x + \frac{b}{2}y + \left(-a + \frac{b}{2}\right)z = 0$ (1)

$(a - b)x + (-a + b)y + (-a + b)z = 0$ (2)

$-\frac{b}{2}x + \frac{b}{2}y + \left(-a + \frac{b}{2}\right)z = 0$ (3)

(1) $-2 \times$ (3): $ax - ay + az = 0 \therefore x - y + z = 0 \therefore y = x + z$

Sub into (2); $(-2a + 2b)z = 0 \ \therefore z = 0$

So $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

When $\lambda_3 = a$ then $\mathbf{A} - b\mathbf{I} = \begin{pmatrix} a - \frac{3b}{2} & \frac{b}{2} & -a + \frac{b}{2} \\ a - b & 0 & -a + b \\ -\frac{b}{2} & \frac{b}{2} & -\frac{b}{2} \end{pmatrix}$

$$\begin{pmatrix} a-\frac{3b}{2} & \frac{b}{2} & -a+\frac{b}{2} \\ a-b & 0 & -a+b \\ -\frac{b}{2} & \frac{b}{2} & -\frac{b}{2} \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow \left(a-\frac{3b}{2}\right)x + \frac{b}{2}y + \left(-a+\frac{b}{2}\right)z = 0$ (1)

$(a-b)x + (-a+b)z = 0$ (2)

$-\frac{b}{2}x + \frac{b}{2}y - \frac{b}{2}z = 0$ (3)

From (2): $x = z$

Sub into (3): $\frac{1}{2}y - z = 0 \therefore y = 2z$

So $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$

$\mathbf{P} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & b \end{pmatrix}$

$\mathbf{D} = \mathbf{P}^{-1}\mathbf{M}\mathbf{P} \Rightarrow \mathbf{P}\mathbf{D} = \mathbf{M}\mathbf{P} \Rightarrow \mathbf{P}\mathbf{D}\mathbf{P}^{-1} = \mathbf{M} \therefore \mathbf{M}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$

$\mathbf{P}^{-1} = \frac{1}{2}\begin{pmatrix} 1 & -1 & 1 \\ 2 & 0 & -2 \\ -1 & 1 & 1 \end{pmatrix}$

$\mathbf{M}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & b \end{pmatrix}^n\begin{pmatrix} 1 & -1 & 1 \\ 2 & 0 & -2 \\ -1 & 1 & 1 \end{pmatrix}$

$= \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 & 0 \\ 0 & a^n & 0 \\ 0 & 0 & b^n \end{pmatrix}\begin{pmatrix} 1 & -1 & 1 \\ 2 & 0 & -2 \\ -1 & 1 & 1 \end{pmatrix}$

$= \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 & 0 \\ 2a^n & 0 & -2a^n \\ -b^n & b^n & b^n \end{pmatrix}$

$= \frac{1}{2}\begin{pmatrix} 2a^n - b^n & b^n & b^n - 2a^n \\ 2a^n - 2b^n & 2b^n & 2b^n - 2a^n \\ -b^n & b^n & b^n \end{pmatrix}$ as req.